

Jaewon Yoon

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EDUCATION

Seoul National University 03 2021 — 02 2027
B.S. in Biological Sciences, Interdisciplinary Major of Artificial Intelligence - (GPA - 3.92/4.3)

COURSEWORK / SKILLS

Cell Biology	Genetics	Biochemistry 1
Systems Biology	Immunology	Introduction to Computer Science for Biologists
Introduction to Bioinformatics	Bioinformatics and Practice 2	
Computer Programming	Data Structure	Algorithm
Linear Algebra 1	Concepts and Applications of Probability	

RESEARCH EXPERIENCE

SNU Laboratory of Computational Biology 12 2024 - 02 2025
Undergraduate Researcher

- Explored the applications of Deep - Learning approaches to transcriptomic data.
- Developed a deep learning model to predict RNA modifications from sequencing data produced by Nanopore sequencing.

SNU Steinegger Lab (Laboratory of Machine Learning and Bioinformatics) 07 2025 - Present
Undergraduate Researcher

- Developed methods for computing significance scores for matches of FoldDisco (a software for protein motif search)
- Currently developing a hybrid protein structure predicting model, combining advantages of different versions of AlphaFold

PUBLICATIONS

Journal paper

- Kim, H., Kim, R.S., Mirdita, M., Yoon, J., Steinegger, M. (2026). Structural motif search across the protein-universe with FoldDisco *bioRxiv*, doi: 10.1101/2025.07.06.663357 (Accepted, Nature Biotechnology)

Others

- Nam, J., Woo, J., Yoon, J. et al., (2025). Unfolding Neuroscience, to be published online.

SCHOLARSHIP / AWARDS

Presidential Science Scholarship — Korea Student Aid Foundation 2021-2026

- 150 undergraduate students selected nationwide for excellency

Encouragement Prize — SNU College, Seoul National University 02 2026

- Awarded to outstanding class founders of the 2025 Student-Directed Education Programs

Dean's list — College of Natural Sciences, Seoul National University Autumn 2021, Autumn 2025

- Awarded to outstanding undergraduate students each semester

Army Commendation Medal — Eighth Army, United States of America 08 2023

- Awarded to winners of General Paik Sun-Yup Leadership Board hosted by the Eighth Army, USFK

Korea Talented Award — Ministry of Education, ROK 11 2020

- 50 high school students and 50 university students selected nationwide

International Biology Olympiad Silver Medal 08 2020

- Scientific competition between international teams of secondary school students

EXTRACURRICULAR - ACADEMIC

Regional Student Group - Korea Core Team, ISCB

01 2026 - Present

Member, Program Committee

- Planned and organized annual symposiums, workshops, and web seminars for RSG-Korea of ISCB
- Invited multiple scholars focusing on antibody structure prediction or omics data analysis for seminars

Deep-learning-based bioinformatics seminar - student-directed education seminar

Autumn 2025

Class founder

- Held weekly student seminars on investigating various fields and researches of computational biology
- Invited multiple scholars focusing on protein structure prediction, protein - ligand interaction prediction, and genomic data analysis for lecture
- Organized presentations and discussions regarding deep-learning based bioinformatics research

Synapse Scholars

07 2024 - Present

Deputy Manager

- Participated in drafting, writing, and publishing the online neurobiology textbook, *Unfolding Neuroscience*
- Wrote or co-wrote the chapters Sensation, Immunology, and Artificial Intelligence

EXTRACURRICULAR - NON-ACADEMIC

SNU College of Natural Sciences Undergraduate Student Mentor (Multi-major)

2025, 2026

Undergraduate Student Mentor

- Provided advice to undergraduate students on selecting and preparing various double majors.

SNU School of Biological Sciences Undergraduate Student Mentor

03 2025 - 02 2026

Undergraduate Student Mentor

- Advised freshman and sophomore students of the department as an undergraduate student mentor.
- Aided overall school life and provided counseling regarding career paths for fellow students.

SNU CNS Science Volunteering Group

2022, 2024-2025

Group member / Member of Group task force - mentoring division

- Participated in mentoring and advising high school students of rural areas nationwide.
- Established and directed multiple programs to academically advise students.

MSC-K HQ, 19th ESC, 8th Army, USFK (Military Service)

08 2022 - 02 2024

KATUSA soldier / Discharged as army sergeant (E-5)

- Served as a Korean soldier augmented to the USFK for mandatory military service of ROK.
- Supported communication and cooperation between USFK stationed in Camp Carroll and Korean army counterparts.

REFERENCES

Prof. Martin Steinegger

Associate Professor, School of Biological Sciences, Seoul National University

E-mail: martin.steinegger@snu.ac.kr

Scholar Profiles: SteineggerLab Homepage — Seoul National University - Personal Page

Prof. Minkyung Baek

Assistant Professor, School of Biological Sciences, Seoul National University

E-mail: minkbaek@snu.ac.kr

Scholar Profiles: Seoul National University - Personal Page

Prof. Milot Mirdita

Assistant Professor, Department of Precision Medicine, Seungkyunkwan University

E-mail: milot@mirdita.org

Scholar Profiles: MirditaLab Homepage